



Module 4: Clustering

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CLUSTERING?

DEFINITION

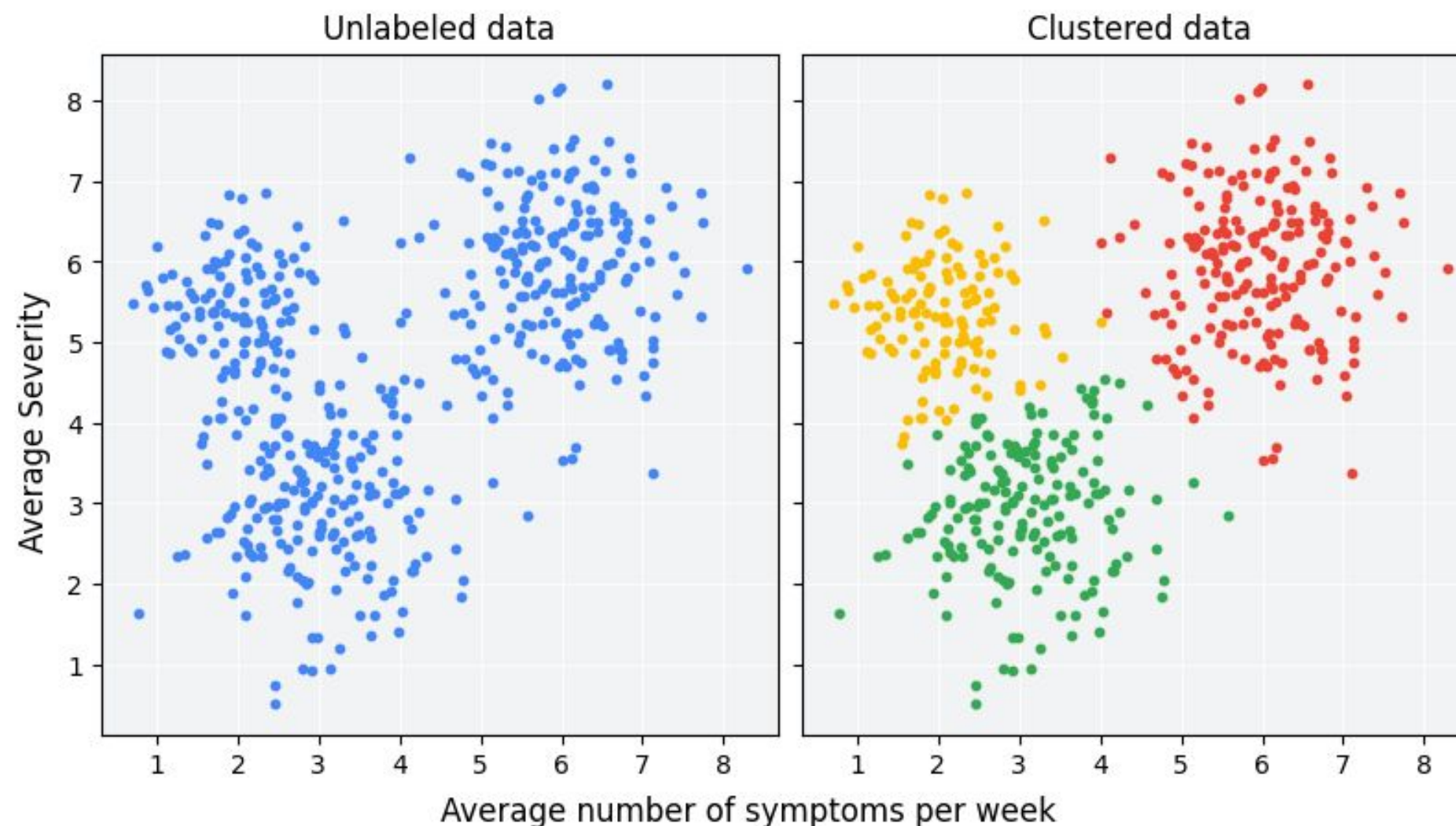
- Clustering is an **unsupervised learning** technique used to group data points.
- Each group (cluster) contains items that are more similar to each other than to items in other groups.
- Unlike classification, clustering doesn't need predefined labels—it discovers patterns automatically.

Example: Similar to stars forming constellations, clustering makes sense of scattered points by revealing natural groupings



GOAL

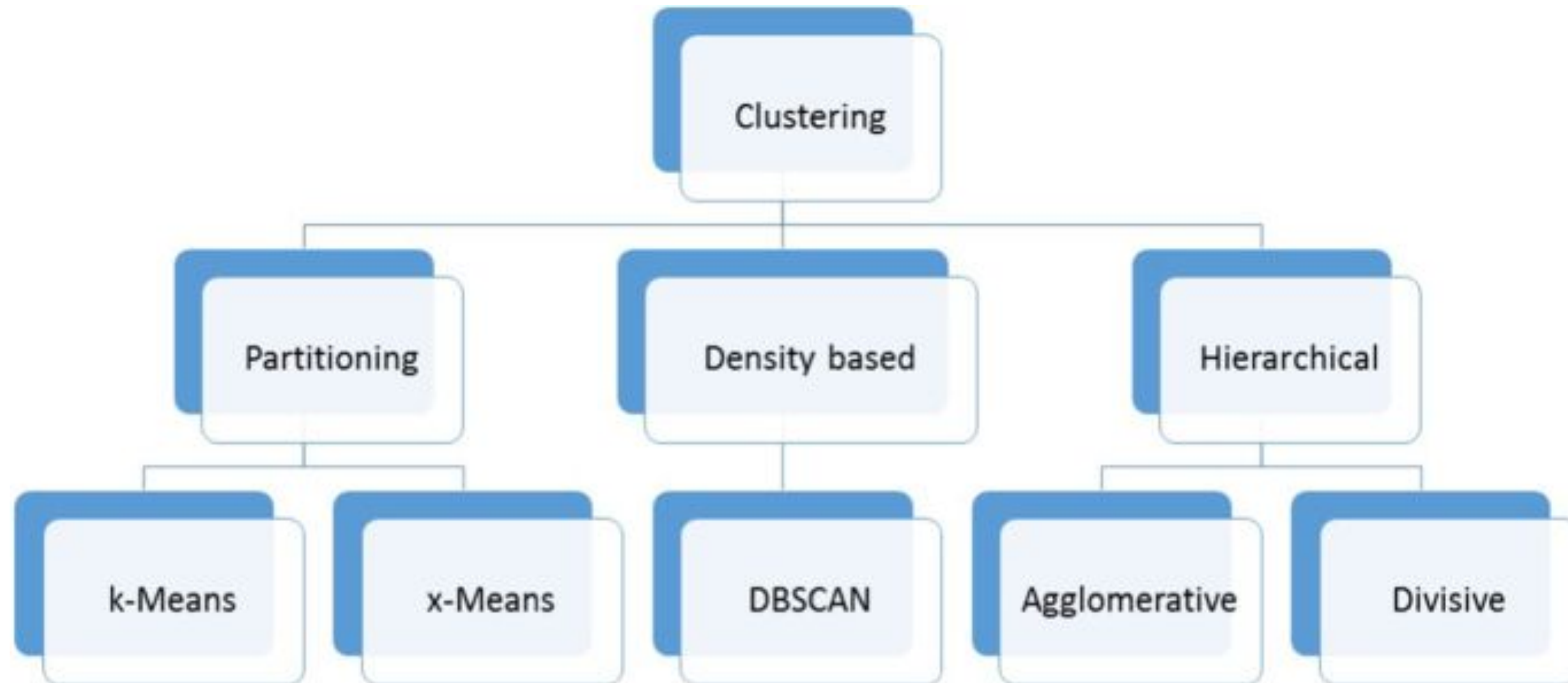
- To identify hidden structures, trends, or relationships in data.
- Helps simplify large datasets by breaking them into meaningful subgroups.
- Often used as a first step in data exploration before applying advanced analytics.



WHY IS CLUSTERING IMPORTANT?

- **Customer Segmentation**
 - Businesses group customers based on buying behavior, demographics, or interests..
- **Image Segmentation**
 - Used in medical imaging, object detection, and facial recognition.
- **Anomaly Detection**
 - Flags unusual patterns that deviate from clusters.
 - Helps in detecting fraudulent credit card transactions or cybersecurity intrusions.
- **Gene Expression Analysis**
 - Groups genes with similar activity patterns.

TYPES OF CLUSTERING ALGORITHMS

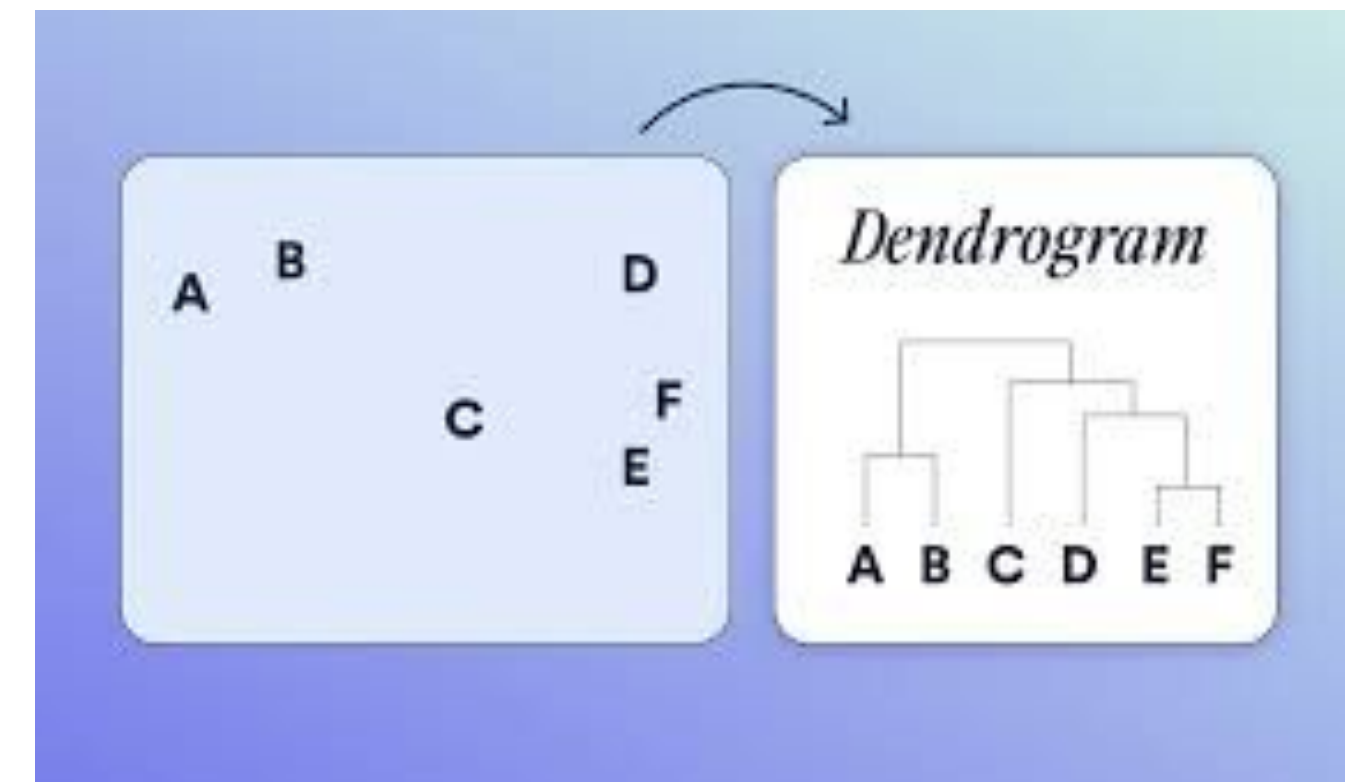


TYPES OF CLUSTERING ALGORITHMS

- **Hierarchical Clustering** → Builds a tree-like structure of clusters.
- **Partitional Clustering** → Divides data into non-overlapping clusters (e.g., K-Means).
- **Density-Based Clustering** → Groups dense areas of data (e.g., DBSCAN Density-Based Spatial Clustering of Applications with Noise).
- **Grid-Based Clustering** → Divides space into grids and clusters them.

HIERARCHICAL CLUSTERING

- Technique that creates a hierarchy of clusters.
- Agglomerative (Bottom-Up) → Start with individual points → merge step by step.
- Divisive (Top-Down) → Start with one big cluster → split step by step.
- Output: Dendrogram (tree diagram showing merging/splitting).
- Does not require pre-specifying the number of clusters K .
- Key Idea: Start with each data point as its own cluster and repeatedly merge the closest clusters.



HOW IT WORKS

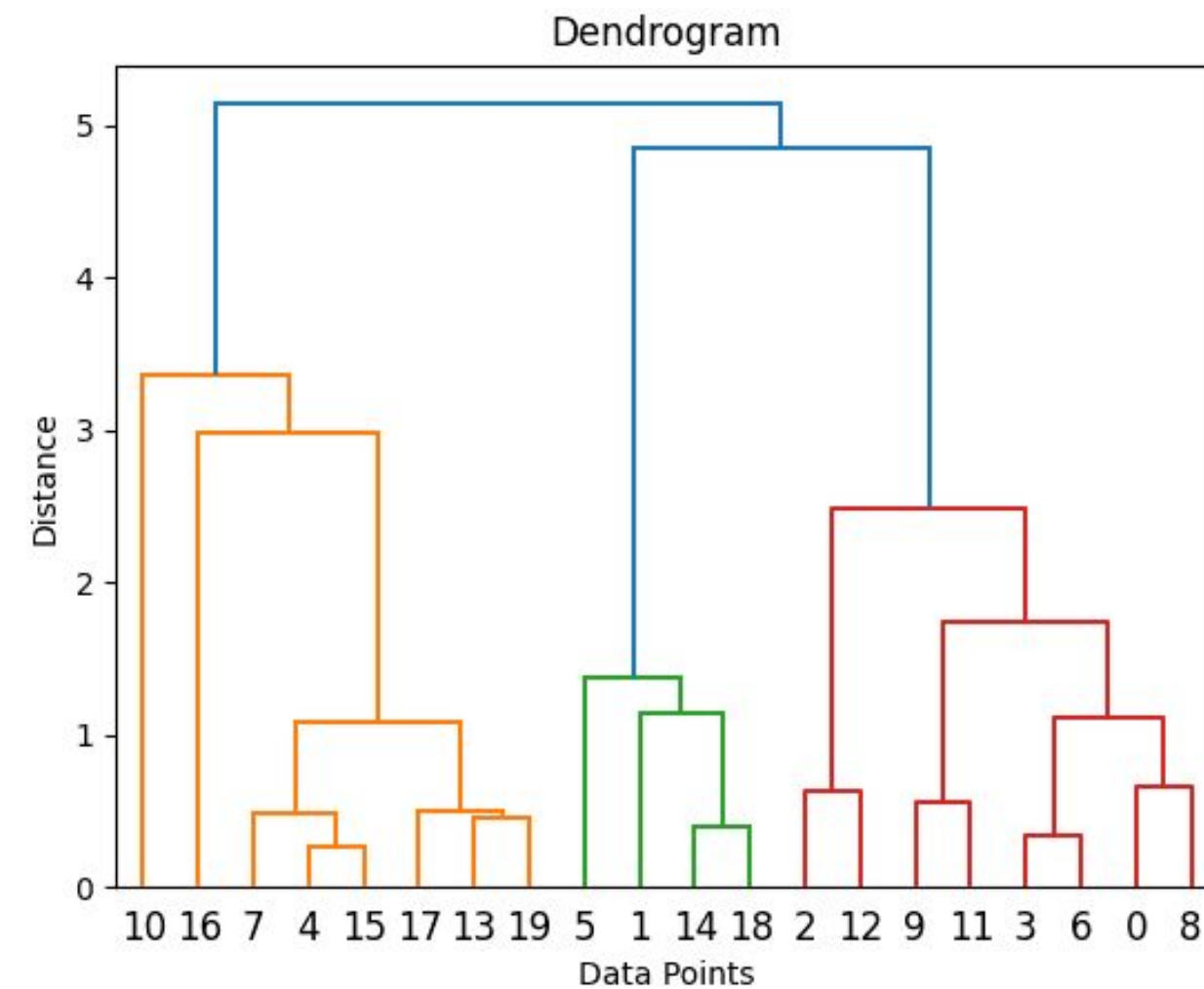
1. Start → Each data point is its own cluster.
2. Find Closest Clusters → Calculate similarity/distance (e.g., Euclidean).
3. Merge → Combine the two closest clusters.
4. Repeat → Continue until only one cluster remains.

HOW DO WE MEASURE DISTANCE BETWEEN CLUSTERS?

- Single Linkage: Distance between the closest pair of points.
- Complete Linkage: Distance between the farthest pair of points.
- Average Linkage: Average distance between all pairs of points.
- Ward's Method: Minimizes the variance within the merged cluster

INTERPRETING THE CLUSTER TREE

- o The height represents the distance at which clusters were merged.
- o To get clusters, "cut" the dendrogram at a chosen height.
- o The number of vertical lines intersected by the cut gives the number of clusters.
- o Advantage: You can choose K after seeing the tree.



Advantages & Disadvantages of Hierarchical clustering

Advantages:-

- No Assumption of Cluster Shape
- Flexible for Categorical Data
- Scalable to Small/Medium Data
- Doesn't Need Multiple Initializations
- Useful in Exploratory Data Analysis (EDA)

Disadvantages:-

- Computationally Expensive
 - Hard to Scale to Big Data
- Noisy/Irrelevant Features Impact Structure
 - Not Robust to Outliers

Partional clustering

Partitional Clustering is a clustering technique in which a dataset is directly divided (partitioned) into k non-overlapping clusters such that each data point belongs to exactly one cluster.

Common Algorithms:

1. K-Means

- Uses centroids (mean of cluster points).
- Minimizes Sum of Squared Errors (SSE).

2. K-Medoids (PAM – Partitioning Around Medoids)

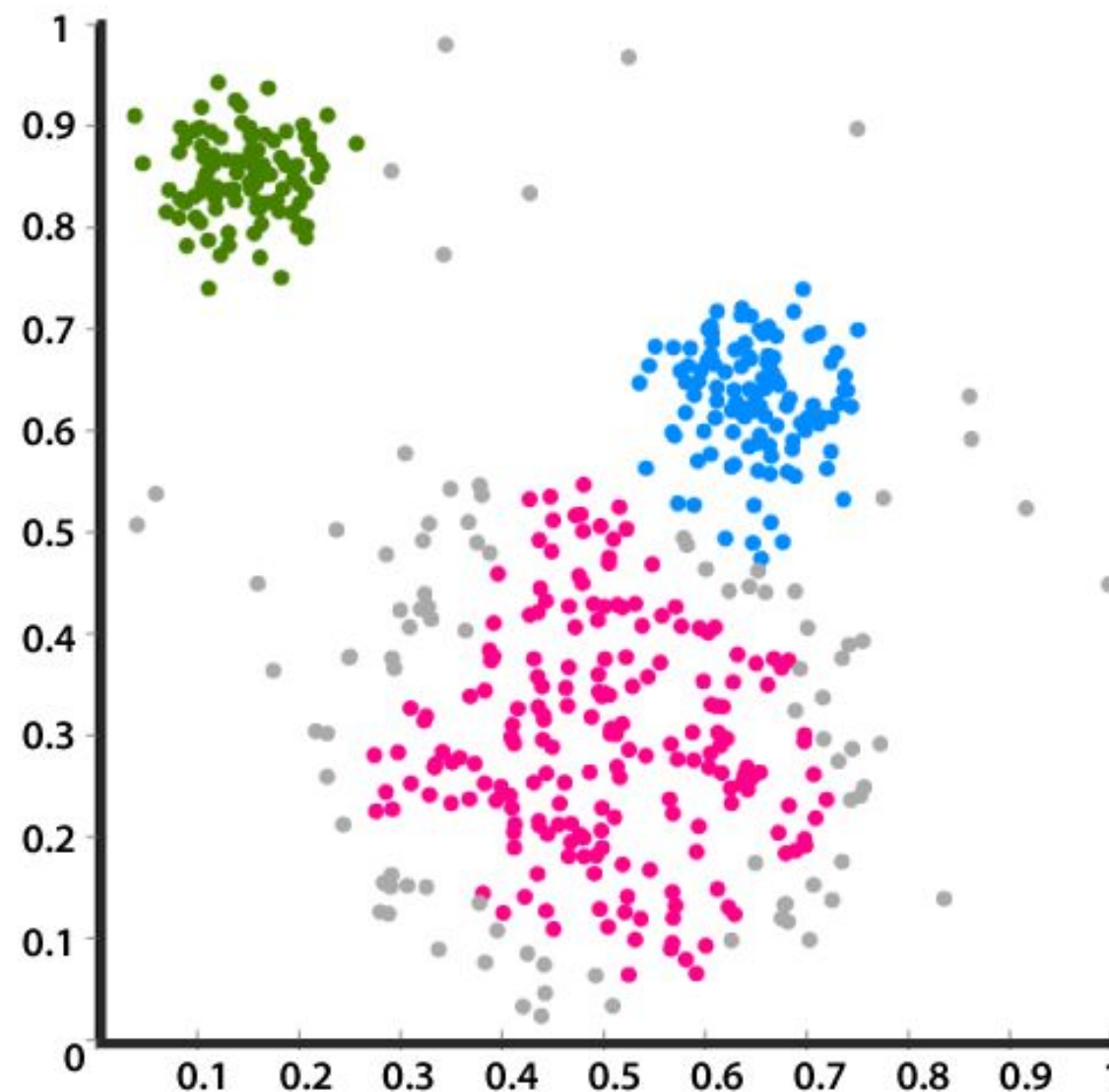
- Uses actual data points (medoids) instead of means.
- More robust to noise and outliers.

3. CLARANS (Clustering Large Applications based on Randomized Search)

- An extension of K-Medoids for large datasets.

K-Means Clustering

K-Means Clustering is an unsupervised machine learning algorithm that helps group data points into clusters based on their inherent similarity. Unlike supervised learning, where we train models using labeled data, K-Means is used when we have data that is not labeled and the goal is to uncover hidden patterns or structures.



K-Means Algorithm

- Choose k → Decide the number of clusters you want to form.
- Initialize centroids → Randomly pick k data points as the initial cluster centers.
- Assign step → For each data point, calculate the distance to all centroids and assign it to the nearest centroid.
- Update step → Recalculate the centroid of each cluster as the mean of all points assigned to that cluster.
- Repeat steps 3–4 until:
 - Centroids stop changing (convergence), or
 - A maximum number of iterations is reached.
- Output → Final clusters and their centroids.

EXAMPLE: Suppose we want to group the visitors to a website using just their age as follows:

16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66

Initial Cluster:

Note: These two points are chosen randomly from the dataset.

K=2
Centroid(C1) = 16 [16]
Centroid(C2) = 22 [22]

Iteration-1:

C1 = 16.33 [16, 16, 17]
C2 = 37.25 [20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66]

Iteration-2:

C1 = 19.55 [16, 16, 17, 20, 20, 21, 21, 22, 23]
C2 = 46.90 [29, 36, 41, 42, 43, 44, 45, 61, 62, 66]

Iteration-3:

C1 = 20.50 [16, 16, 17, 20, 20, 21, 21, 22, 23, 29]
C2 = 48.89 [36, 41, 42, 43, 44, 45, 61, 62, 66]

Iteration-4:

C1 = 20.50 [16, 16, 17, 20, 20, 21, 21, 22, 23, 29]
C2 = 48.89 [36, 41, 42, 43, 44, 45, 61, 62, 66]

No change Between Iteration 3 and 4, so we stop. Therefore we get the clusters (16-29) and (36-66) as 2 clusters we get using K Mean Algorithm.